W-G-656D
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SUPERSEDING
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May 12, 1967 and
Fed. Spec. W-G-656B
March 31, 1958

FEDERAL SPECIFICATION

GRINDING MACHINE, AND GRINDING AND BUFFING MACHINE UTILITY, BENCH AND PEDESTAL, ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric motor driven, utility type, grinding machines and grinding and buffing machines with equipment and accessories as described herein.

1.1.1 Federal specification coverage. Federal specifications do not cover all types and sizes of the commodity indicated by the titles of the specifications, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. Grinding machines and grinding and buffing machines covered by this specification shall be of the following types and sizes as specified (see 6.2):

Type I - Grinding machines, bench, two-wheel, dry grind.

Type II - Grinding machines, pedestal, two-wheel, dry grind.

Size	Grinding Wheel Diameter
6	6 inch
7	7 inch
8	8 inch
10	10 inch
12	12 inch
14	14 inch

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Type III - Grinding machines, pedestal, two-wheel, combination wet-dry.

Size	Grinding Wheel Diameter
12	12 inch

Type IV - Grinding and buffing machines, bench.

Type V - Grinding and buffing machines, pedestal.

Size	Grinding or Buffing Wheel Diameter
8	8 inch
10	10 inch
12	12 inch
14	14 inch

When other sizes are required, the machine size and its corresponding characteristics shall be as specified (see Table I and 6.6).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

J-C-175 - Cable Assembly, Power, Electrical (3 wire, 3 prong, Grounding Plug Connector, For 125-Volt Equipment).
W-C-596 - Connector, Plug Electrical; Connector, Receptacle, Electrical.
CC-M-636 - Motor, Alternating-Current, (Fractional Horsepower).
CC-M-641 - Motor, Alternating Current, (Integral Horsepower, 200 HP and Smaller).
CC-M-645 - Motor, Direct Current (Fractional-Horsepower).
CC-M-646 - Motor, Direct Current (Integral-Horsepower, 200 HP and Smaller).
GGG-C-350 - Chucks, Drill, Arbors and Keys, Drill Chuck.
GGG-D-631 - Dresser, Abrasive Wheel, Hand, Revolving Cutter Type.

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116	-	Preservation, Methods of.
MIL-W-17929	-	Wheels, Abrasive, Precision.
MIL-M-18058	-	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Occupational Safety and Health Act (OSHA) of 1970:

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Underwriters' Laboratories (UL), Inc., Standards:

UL - 45 - Standards for Electric Tools.
UL - 987 - Stationary and Fixed Electric Tools.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, NY 11749; 207 East Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Electrical Manufacturers' Association (NEMA) Standards:

MG-1 - Motors and Generators.
IC-1 - Industrial Controls.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

American National Standards Institute (ANSI), Inc., Standards:

B5.17	-	Marking for Grinding Wheels.
B7.1	-	American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels.
B74.2	-	Shapes and Sizes of Grinding Wheels.

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the product are classified herein as follows:

Requirement	Paragraph
First Article approval	3.2
General requirements	3.3
Detail requirements, Type I	3.4
Detail requirements, Type II	3.5
Detail requirements, Type III	3.6
Detail requirements, Type IV	3.7
Detail requirements, Type V	3.8

3.2 First Article approval. If First Article approval is required (see 6.2 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

3.3 General requirements.

3.3.1 Design. The machine shall be new and one of the manufacturer's latest designs. Machine design shall comply with the requirements herein in accordance with the latest state-of-the-art. Each machine shall be capable of off-hand grinding or grinding and buffing of numerous tools and work parts to varying finishes and configurations. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repairs.

3.3.2 Controls. The machine shall be completely wired through the controls. The controls shall conform to NEMA IC-1 and shall be installed in a NEMA Type I enclosure or better. Unless otherwise specified (see 6.2), the ON-OFF switch, on machines with 2 or more horsepower, shall incorporate overload protection if not provided in the motor or control. All controls shall be readily accessible to the operator.

3.3.3 Safety measures. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can

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be guarded without substantial interference to operation of the machine. All guards shall be removable to provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.3.4 Lubrication. All bearings (except sealed-for life-type) shall be provided with accessible means of lubrication. All oil holes shall have filler caps of the self-closing type to prevent grit or dust from entering.

3.3.5 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

3.3.6 Construction. The machine shall be so constructed that when installed and connected to power, it shall be ready for operation. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.7 Castings and forgings. All castings and forgings shall be substantially free from blowholes, porosity, hard-spots, shrinkage defects, cracks, or other injurious defects. In no event shall such processes as welding, peening or plugging be used on castings or forgings for reclaiming any part of the machine without the authorization of the procuring activity.

3.3.8 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.9 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.10 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise

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specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

3.3.11 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.12 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.4.1 Design. Type I machines shall consist essentially of a base, grinding wheel assembly, wheel hoods or guards, workrests, eye shields, ON-OFF switch, grinding wheels, and when applicable (see 3.4.4), an exhaust outlet connection.

3.4.2 Base. The base shall be of metallic composition, fabricated or cast of iron, steel or aluminum, arranged to support the grinding wheel assembly and all other associated components of the machine. The ON-OFF switch shall be housed in the base convenient to the operator. Means for anchoring the base to a bench shall be provided. The base shall be provided with a bottom cover plate to shield it from the entrance of dust or other foreign matter. Base dimensional characteristics shall be manufacturer's standard.

3.4.3 Grinding wheel assembly. The grinding wheel assembly shall include the work spindle and its related components, spindle housing, drive mechanism, and grinding wheels and wheel flanges. Each work spindle shall be of high quality steel with a minimum tensile strength of 90,000 lbs. per square inch (psi), and a minimum hardness of 24 Rockwell "C". The spindle shall be designed to receive a grinding wheel at each end. Each work spindle shall be mounted on ball or roller bearings and shall be accurately machined so radial runout does not exceed 0.005 inch total indicator reading (TIR). Spindles and motor bearings shall be protected from dirt, swarf, coolant and other foreign matter by seals or shields. Unless otherwise specified (see 6.2), the motor shaft shall be used as a work spindle. Motor shafts shall operate in lubricated-for-life-type bearings, installed with means for minimizing end play. Motor control shall be through a switch having two marked positions: ON and OFF. Means for securing the wheels and wheel flanges to the work spindle shall be provided.

3.4.4 Hoods, guards, flanges, and spark arrestors. Each machine shall be provided with a hood or guard fitted over each grinding wheel. Wheel flanges shall be provided for each grinding wheel. The hoods or guards and flanges shall be in accordance with ANSI B7.1. Each hood or wheel guard shall be provided with a spark arrestor.

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Hoods or guards shall provide easy accessibility for grinding wheel changes. Sizes 10 inch, 12 inch, and 14 inch machine hoods or guards shall have an exhaust outlet connection. When specified (see 6.2), sizes 7 inch and 8 inch machine hoods or guards shall have an exhaust outlet connection.

3.4.5 Eye shields. Unless otherwise specified (see 6.2), an adjustable eye shield shall be provided for each grinding wheel. Each eye shield shall have a transparent pane made of shatterproof glass or high impact resistant transparent thermosplastic material. Each eye shield shall be so designed and mounted that it can be adjusted and locked in any position, or completely removed. When specified (see 6.2), each eye shield shall be furnished with a work light.

3.4.6 Workrests. A workrest shall be provided for each grinding wheel. Workrests shall be of metallic composition, fabricated or cast of iron, steel or aluminum, and shall conform to ANSI B7.1. Workrests shall be adjustable.

3.4.7 Grinding wheels. Grinding wheels shall be in accordance with MIL-W-17929. Wheel shapes shall conform to ANSI B74.2. Grinding wheels shall be marked in accordance with ANSI B5.17, and shall meet the safety requirements designated in ANSI B7.1.

3.5 Detail requirements, Type II grinding machine, pedestal, two-wheel.

3.5.1 Design. Type II machines shall consist essentially of the same basic components as Type I machines (see 3.4 through 3.4.8), except a pedestal shall be provided in lieu of the bench type base. The pedestal shall be of metallic composition, fabricated or cast of iron, steel or aluminum, and designed for floor mounting. The structure shall be arranged to support or enclose the grinding wheel assembly and all other associated components of the machine. When applicable, door(s) or readily removable cover(s), or plate(s), shall be provided for servicing any internally housed components. A water cup, for cooling work parts, shall be provided in an easily accessible location. Means for anchoring the pedestal's base to the floor shall be provided. When specified (see 6.2), the pedestal shall be provided with a workholding tray. The pedestal's base dimensional characteristics shall be manufacturer's standard.

3.6 Detail requirements, Type III grinding machine, pedestal, combination wet-dry.

3.6.1 Design. The Type III machine shall be constructed basically the same as Types I and II machines (see 3.4 through 3.5.1), utilizing

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a pedestal and its other components; and also shall incorporate a built-in type coolant system for wet grinding. The machine shall be arranged with a dry type grinding wheel on one end of the work spindle and a suitable wet grinding abrasive wheel on the opposite end. The wet grinding side of the machine shall have a splash bowl, water tank with settling chamber and pumps, water piping, and an adjustable nozzle and valve. The splash bowl and tank shall be of metal and be securely attached to the machine. The tank shall be readily accessible for clean-out purposes or replacement, and shall have a reservoir capacity of not less than 2-1/2 gallons. The pump shall be so arranged and attached that its bearings cannot be submerged in water. The dry grinding side of the machine shall be provided with an exhaust outlet connection in accordance with 3.4.4.

3.7 Detail requirements, Type IV grinding and buffing machine, bench.

3.7.1 Design. The Type IV machine shall consist essentially of a base, grinding and buffing wheel assembly, grinding wheel hood or guard, workrest, eye shield, ON-OFF switch, grinding wheel, cloth or wire wheel, and when applicable (see 3.4.4), an exhaust outlet connector.

3.7.2 Base. The base shall be of metallic composition, fabricated or cast of iron, steel or aluminum, and arranged to support the grinding and buffing wheel assembly and all other associated components. The ON-OFF switch shall be housed in the base convenient to the operator. Means for anchoring the base to a bench shall be provided. A bottom cover plate shall be provided to shield the base from the entrance of dust or other foreign matter. Base dimensions shall be manufacturer's standard.

3.7.3 Grinding and buffing wheel assembly. The grinding and buffing wheel assembly shall include the work spindle and its related components, spindle housing, drive mechanism, grinding wheel, cloth or wire wheel and wheel flanges. The spindle and spindle housing on the buffing wheel side shall be of an elongated configuration to allow the operator more working freedom. Work spindles shall be of high quality steel with a minimum tensile strength of 90,000 psi and a minimum hardness of 25 Rockwell "C". The spindle shaft shall be designed to accept a grinding wheel on one end and a buffing wheel (cloth or wire) on the other end. The work spindle shall be mounted on not less than three ball or roller bearings, with at least one bearing located in the extreme end of the spindle housing at the buffer end. The work spindle shall be accurately machined so radial runout does not exceed 0.005 inch (TIR). Spindles 1 inch diameter and larger shall be tapped to accept a spindle extension. The

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work spindle and motor bearings shall be protected from dust, swarf, coolant and other foreign matter by seals or shields. Unless otherwise specified (see 6.2), the motor shaft shall be used as a work spindle. Motor shafts shall operate in lubricated-for-life-type bearings installed with means for minimizing end play. Motor control shall be through a switch having two marked positions: ON-OFF. Means for securing wheels and wheel flanges to the work spindle shall be provided. The requirements for hoods or guards, flanges, spark arrestor, eye shield, workrest, and grinding wheel(s) contained in 3.4.4 through 3.4.7 are also applicable to Type IV machines.

3.8 Detail requirements, Type V grinding and buffing machine, pedestal.

3.8.1 Design. The Type V machine shall consist essentially of the same basic components as Type IV machines (see 3.7 through 3.7.3), except a pedestal shall be provided in lieu of the bench type base. Pedestal requirements in 3.5.1 for Type II machines are applicable to Type V machines.

3.9 Each machine shall meet the requirements of Table 1.

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TABLE I

Sizes and Capacities

CHARACTERISTIC	** REQUIREMENT										
	TYPE I AND TYPE II						TYPE III	TYPE IV AND TYPE V			
	Size 6	Size 7	Size 8	Size 10	Size 12	Size 14	Size 12	Size 8	Size 10	Size 12	Size 14
Grinding wheel (abrasive)											
diameter, inches	6	7	8	10	12	14	12	8	10	12	14
face, inches	5/8	3/4	7/8	1	2	2-1/2	2	1	1	2	2-1/2
distance between wheels, inches	9-1/4	13-3/16	14	16-9/16	20-1/2	21	22-3/4	-	-	-	-
Buffing wheel (cloth or wire)											
diameter, inches	-	-	-	-	-	-	-	8	10	12	14
face, inches	-	-	-	-	-	-	-	1	1-1/4	2	2-1/2
distance between wheels, inches	-	-	-	-	-	-	-	22-1/4	24-9/16	29	29
Buffing spindle extension, inches	-	-	-	-	-	-	-	12	12	16	16
Height, base to spindle centerline, inches, pedestal type	36	36	36	36	36	36	36	36	36	36	36
bench type	4-1/4	4-7/8	5-1/2	6-1/2	14	15-1/4	-	5-1/2	6-1/2	14	15-1/4
Spindle speed, number	1	1	1	1	1	1	1	1	1	1	1
Spindle speed, revolutions per minute, (RPM)	3450	3450	3450	1725	1725	1725	1725	3450	1725	1725	1725
Spindle shaft diameter, inches	1/2	1/2	5/8	3/4	1-1/4	1-1/4	1-1/4	1 5/8	1-1/4	1-1/4	1-1/4
Spindle drive motor, horsepower	1/3	1/3	3/4	1	2	3	2	3/4	1	2	3
Coolant pump motor, horsepower	-	-	-	-	-	-	1/10	-	-	-	-

** Unless otherwise specified (see 6.2).

Requirements are not less than those shown.

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3.10 Accuracy and performance. Each machine shall be capable of off-hand grinding operations as associated with normal shop practice. Each machine shall be capable of passing the tests in 4.4.1 and 4.4.2.

3.11 Electrical equipment. Unless otherwise specified (see 6.2), the electrical system shall conform to the applicable standards of the National Electrical Manufacturers' Association, and Underwriters' Laboratories, Incorporated, UL-45 and 987.

3.11.1 Motor. Motors shall be of the continuous duty, totally enclosed type. Alternating current motors shall conform to Federal Specification CC-M-636 or CC-M-641, as applicable, except for fungus control and radio interference requirements as herein specified. Single-phase motors shall be of the repulsion start or capacitor start type. Three-phase motors shall be of the squirrel-cage induction type. Direct current motors shall be of the compound-wound or shunt-wound type and shall conform to Federal Specification CC-M-645 or CC-M-646, as applicable, except for fungus control and radio interference requirements as herein specified. Motors shall be of the type specified (see 6.2).

3.11.1.1 Motor nameplate. Each motor shall be equipped with the motor manufacturer's nameplate bearing the motor manufacturer's name, the motor model number, the serial number, rated horsepower, speed in revolutions per minute, and the motor electrical characteristics. With the motor mounted on the machine, the nameplate shall be easily visible without removing or turning the motor.

3.11.2 Motor rating. Unless otherwise specified (see 6.2), the motor for each grinding machine and grinding and buffing machine shall be of the rating specified in Table I.

3.11.3 Power supply. Power supply characteristics shall be as specified (see 6.2).

3.11.4 Starting mechanism. Each machine shall be provided with a switch, of the proper capacity, for the motor horsepower used. The switch case shall be of suitable insulated material or, if of metal, it shall be suitably insulated from current carrying parts and effectively grounded. Machines of 1-horsepower and over shall be equipped with a push button and magnetic starter switch having overload protection and low voltage release; machines of less than 1-horsepower shall be provided with a built-in switch and overload protection. A push button and magnetic starter shall be furnished on all machines having over 220 volts. The switch shall be located

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convenient to the operator. Direct current motors of 1-horsepower or more shall be provided with time limit acceleration and shunt field failure protection. Fractional horsepower DC motors shall be provided with overload switch. When specified (see 6.2), the machines shall be provided with dustproof switch.

3.11.5 Cable and plug. Each machine using a single-phase AC or DC motor shall be furnished with not less than 6-feet of 3-conductor type cable conforming to Federal Specifications J-C-175, Type 1 cable assemblies and W-C-596, Style D, series 2-pole, 3-wire grounding type, 15-ampere, 125-volt. Each machine using 3-phase, 220-volt, AC motor shall be provided with a junction box enclosing 4 terminals (one grounded to machine frame). Terminals shall be of a size sufficient to carry continuously full rated current. The three-prong polarized connector plug shall be provided with mating three to two parallel prong adapter with grounding lead for use with a two wire outlet.

3.12 Standard equipment. Unless otherwise specified (see 6.2), the following equipment shall be furnished with each machine:

(a) Type I, II, and III machines, two grinding wheels of the rated size (see 1.2 and Table I). The material composition (grain, grit, and bond) shall be as specified.

(b) Type IV and V machines, one each, grinding wheel, cloth wheel, and wire wheel of the rated size (see 1.2 and Table I). The material composition (grain, grit, and bond) and other applicable characteristics for wheels shall be as specified.

3.13 Optional equipment. When specified (see 6.2), the following optional equipment shall be furnished:

3.13.1 Drill grinding attachment. The drill grinding attachment, when specified for this type of grinding machine, shall be a detachable drill grinding attachment suitable for accurately grinding drill point angles from 90° to 130° on twist drills, or flat drills having straight or taper shanks. The attachment shall be furnished complete with all necessary adapters or equipment required to mount it in place. It shall be adaptable and suitable for the size and model of the grinding machine furnished.

3.13.2 Edge tool grinding attachment. When the edge tool grinding attachment is specified (see 6.2), it shall be provided with a removable tool holder. The attachment shall have a micrometer screw feed adjustment, and shall be capable of grinding plane bits up to and including 2-5/8 inches wide and flat woodworking chisels of any size.

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3.13.3 Rock bit grinding attachment. The rock drill grinding attachment, when specified (see 6.2), shall be detachable and suitable for accurately grinding 4-point cross rock drill bits with reverse type buttress threads in sizes up to 3 inches. The attachment shall be furnished complete with all necessary adapters or equipment required to mount it in place. It shall be adaptable and suitable for the size and model of the machine furnished.

3.13.4 Drill chucks. When drill chucks are specified (see 6.2), they shall conform to Type I, Class 2 of Specification GGG-C-350 and shall be provided with a threaded mounting hole of a size and thread to fit the threaded ends of the spindle for the size and model machine furnished.

3.13.5 Dressers, abrasive wheels. Dressers shall be of the diamond point type or the revolving cutter type as specified (see 6.2). The revolving cutter type shall conform to Federal Specification GGG-D-631 and be supplied with an extra cutter set.

3.14 Electromagnetic interference control. When specified (see 6.2), equipment procured under this specification shall comply with MIL-STD-461 for Class IIB equipment.

3.15 Fungus control. When required (see 6.2), fungus proofing shall be as specified.

3.16 Lubrication chart or plate. When specified (see 6.2), lubrication chart or plate shall be securely attached to the machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application.

Service interval.

Type of lubricant.

Viscosity.

Federal or Military Specification Number (if available).

3.17 Nameplate. When specified (see 6.2), a corrosion-resistant nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each machine. If the machine is a special model, the model

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designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records. The captions may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature.

Manufacturer's name.

Manufacturer's model designation.

Manufacturer's serial number.

Size _____.

Grinding or buffing wheel size _____ inch diameter _____ inch face _____ inch arbor hole.

Power input (volts, total amps, phase, frequency).

Contract Number or Order Number.

Federal Stock Number or Plant Equipment Code.

Date of manufacture.

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3.18 Identification plate. When specified (see 6.2), a separate identification plate of corrosion-resistant metal permanently and legibly marked in letters 3/16 inch high with the caption "Departmental or Activity Identification" shall be securely attached to each machine. The plate shall have blank space for the Government to stamp 25 characters of the same size as the caption.

3.19 Technical data.

3.19.1 Civil Agency Procurements. For procurements by Civil Agencies, two copies each of the Manufacturer's Handbook of Instructions for Machine Operation, Care, and Maintenance, a machine Parts List, and two copies each of Foundation and Anchor Bolt Plans and Wiring Diagrams shall be furnished.

3.19.2 Military Procurements. Data shall be furnished as specified (see 6.2.2).

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3.20 Workmanship. Workmanship of the machine and accessories shall be of the quality prevailing among manufacturers normally producing machines of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.2, First Article inspection shall be performed. Unless otherwise specified (see 6.2), First Article inspection shall comprise examination for compliance with the applicable requirements in 3.3 through 3.20, examination to insure compliance with the additional safety and health requirements specified by the procuring activity under 6.2.1(e), and all tests in 4.4. Failure to pass any examination or any test shall be cause for disapproval of the First Article.

4.3 Inspection for acceptance. Each machine shall be examined in accordance with 4.3.2 and tested in accordance with 4.4.1. Unless otherwise specified (see 6.2), additional inspection shall be made in accordance with the provisions set forth in MIL-STD-105, as specified in 4.3.3, 4.3.4, and 4.3.5. The inspection terms and definitions shall be interpreted in accordance with MIL-STD-109. Results shall be kept available for review by Government inspectors.

4.3.1 Inspection of materials and components. The supplier shall ascertain that components and materials used were manufactured, tested, and inspected to insure that machines can be produced in accordance herewith.

4.3.2 Examination for safety and health requirements. Each machine shall be examined to insure compliance with 3.3.3 and the additional safety and health requirements specified by the procuring activity under 6.2.1(e) (see 6.5). Each machine shall comply with all such requirements.

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4.3.3 Examination of size and capacity. Examination shall be made of each end item in the sample to determine compliance with the requirements of Table I. Any characteristic which does not comply with Table I shall be classified as a defect. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

4.3.4 End product visual examination. Each completely assembled end item in the sample shall be examined for defects in finish, material, construction, workmanship and marking. Defects shall be classified as outlined in Table II for recording of inspection results and corrective action. Inspection level shall be S-2 with an AQL of 4.0 defects per hundred units for major defects, and 6.5 defects per hundred units for minor defects.

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TABLE II. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Type of finish not as specified or blistered, peeled, chipped, or thin film.		X
	Not finished where required.	X	
	Component or assembly that is painted or otherwise coated where serviceability may be affected, e.g., paint on gasket.	X	
Material	Component not fabricated of the specified material.	X	
Design	Any characteristic not in accordance with the specified requirements.	X	
Construction and workmanship, general (applicable to all components and assemblies)	Part missing or not specified type of size.	X	
	Fractured, split, punctured, dented, deteriorated, bowed, malformed, or otherwise imparied affecting serviceability.	X	
	Defective or otherwise damaged but not affecting serviceability.		X
	Misplaced, loose, or not in proper alignment affecting serviceability.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel or may affect serviceability.	X	
	Operation omitted or not properly performed, affecting serviceability.	X	
	Operation omitted or not properly performed, but not affecting serviceability.		X
	Functioning component that requires abnormal force to operate.		X
	Not connected or joined as specified or poorly accomplished, affecting serviceability,	X	

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TABLE II. Classification of defects (Cont'd)

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship (con.)	Any adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended.	X	
	Any component not readily accessible for servicing where required.		X
Welding and brazing (where required)	Not type specified, missing, incomplete, burn holes, cracked, fractured, otherwise not fused.	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed.		X
Soldering (when applicable)	Missing, not adherent or incomplete.	X	
	Not clean (flux or flux residue not removed) not smooth (surface not finished neatly) or pin holes in solder.		X
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, broken, stripped, fractured, or loose.	X	
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal or not inclosed at specified conduit where required.	X	
	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring.		X
Lubrication (where applicable)	Not lubricated where required.	X	
	Insufficiently lubricated.		X
	Not specified type lubrication fitting, affecting serviceability.	X	
	Lubrication fitting not readily accessible for servicing.		X

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TABLE II. Classification of defects (Cont'd)

Examine	Defect	Classification	
		Major	Minor
Assembly	Component not properly assembled or secured but not affecting serviceability; e.g., lockwasher missing from metal fastener.		X
	Unit perceptibly out of square or alignment.	X	
Marking identification, instructions, manual	Missing, incomplete, not legible		X

4.3.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table III. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery and selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects.

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of container.

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4.4 Tests.

4.4.1 Operational test. Each machine shall be operated at no load for not less than 5 minutes, including at least three starts. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be proved during the trial period. The machine shall accomplish each start properly, and shall run smoothly without excessive vibration.

4.4.2 Upsetting moment test, pedestal type machines only. The fully assembled First Article unit shall be tested for upsetting by exerting a force equal to one-tenth the gross weight of the unit to the grinding wheel. The direction of the force shall be parallel to the floor and perpendicular to the spindle. The machine shall be blocked at the floor to prevent sliding but not anchored to prevent upsetting. A spring scale secured to a cord around the wheel and spindle may be used to apply the force. The base shall remain firmly on the floor.

4.4.3 Electromagnetic interference control tests. When electromagnetic interference control is required, the First Article shall be tested for compliance with 3.14 in accordance with Table II of MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2).

5.1.1 Technical publications. Technical publications for each machine shall be preserved in accordance with MIL-P-116, Method IC-1.

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B, or C, as specified (see 6.2).

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order (see 6.2), the interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order (see 6.2), the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The grinding machines and grinding and buffing machines covered by this specification are for general grinding and buffing operations.

6.2 Ordering data. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.
(b) Type and size machine required (see 1.2).
(c) First Article approval, if required (see 3.2).
(d) Controls, if different (see 3.3.2).
(e) Specify exceptions and additional safety and health requirements (see 3.3.3 and 6.5).
(f) Painting, if different (see 3.3.11).
(g) Spindle drive, if different (see 3.4.3 and 3.7.3).
(h) Specify if size 7 or 8 machines require exhaust outlet (see 3.4.4).
(i) Specify if eye shields are not required (see 3.4.5).
(j) Specify if work light is required (see 3.4.5).
(k) Specify if workholding tray is required (see 3.5.1).
(l) Machine requirements, if different (see Table I).
(m) Electrical system, if different (see 3.11).
(n) Specify motor type required (see 3.11.1).
(o) Motor rating, if different (see 3.11.2).
(p) Power supply characteristics required (see 3.11.3).
(q) Specify if dustproof switch is required (see 3.11.4).
(r) Standard equipment, if different (see 3.12).
(s) Optional equipment, if required (see 3.13).
(t) Electromagnetic interference control, if required (see 3.14 and 6.4).
(u) Type fungus proofing, if required (see 3.15).
(v) Lubrication chart, if required (see 3.16).
(w) Nameplate, if required (see 3.17).
(x) Identification plate, if required (see 3.18).
(y) First Article inspection, if different (see 4.2).
(z) Inspection for acceptance, if different (see 4.3).
(aa) Required level of preservation-packaging (see 5.1).
(bb) Required level of packing (see 5.2).
(cc) Additional marking, if required (see 5.3).

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6.2.2 Contract data requirements. For military procurements, required technical data, such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.19).

6.3 First Article. When First Article approval is required under 3.1, the solicitation should include complete information concerning examination, testing, and approval.

6.4 Army and Air Force procurements. When equipment purchased for the Departments of the Army and the Air Force requires electromagnetic interference control testing, such testing should be specified in accordance with MIL-STD-461, Class IIB equipment, except only Test Methods CE03 and RE02 should be required.

6.5 Occupational Safety and Health Act (OSHA). Paragraph 3.3.3 requires compliance only with OSHA requirements concerning the machine itself. These requirements include, but are not limited to, the following paragraphs of the OSHA Standards:

1910.144 - Safety color code for marking physical hazards

1910.212 - General requirements for all machines

1910.215 - Abrasive wheel machinery

1910.219 - Mechanical power transmission apparatus (when applicable)

Since this specification does not require compliance with OSHA requirements concerning the operating environment, procuring agencies should insert additional requirements under 6.2.1(e) that will integrate the new machine into the work environment without exceeding OSHA hazard levels. OSHA paragraphs applicable to this specification which are pertinent to the operating environment are:

1910.93 - Air contaminants

1910.95 - Occupational noise exposure

The above paragraphs are cited to assist procuring agencies in complying with OSHA. The machine user should measure and evaluate his safety and health requirements based on the entire contents of OSHA standards.

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6.6 Size identification. The machines covered by this specification are sized in accordance with nominal grinding or buffing wheel outside diameter.

MILITARY CUSTODIANS:

Army - WC
Navy - SH
Air Force - 84

Review activities:

Army - WC, ME
Navy - SH
Air Force - 84

User activities:

Navy - MC, YD, OS
DSA - GS

Preparing activity:

DSA-IP

Project number:

3415-0106

CIVIL AGENCIES COORDINATING ACTIVITY:

HEW - FDA
GSA - FSS

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